

Revision Notes for Class 5 Maths Chapter 2 Fractions

In this chapter, you will learn about fractions, which are a way to represent parts of a whole. Fractions are important because they help us relate numbers to real-life concepts like sharing food, dividing objects, and measuring. Understanding how to compare, add, and find equivalent fractions is the main focus here.

Understanding Fractions and Wholes

Fractions represent a number of equal parts of a whole object. For example, if a chocolate is divided into three equal pieces, each piece is called one-third ($\frac{1}{3}$). To compare fractions like $\frac{1}{2}$ and $\frac{1}{3}$, it's important that the "wholes" (the original objects being divided) are the same size, otherwise the comparison may not make sense. For instance, $\frac{1}{3}$ of a larger chocolate can be more than $\frac{1}{2}$ of a smaller one.

Playing with Grids and Fraction Kits

Practical activities such as shading grids and using fraction kits help students visualize fractions. For example:

- Shade $\frac{1}{8}$ of Grid A in red.
- Shade $\frac{1}{6}$ of Grid B in blue.
- Shade $\frac{1}{12}$ of Grid C in yellow.

These activities encourage finding $\frac{1}{2}$ or $\frac{1}{3}$ in different shapes and help you see how fractions look visually in parts of grids or objects.

Using a fraction kit, students can combine pieces to make a whole. For example, one piece of $\frac{1}{2}$ and two pieces of $\frac{1}{4}$ together form a whole. This shows that $\frac{1}{2} = \frac{2}{4}$, demonstrating the concept of equivalent fractions. When you divide a $\frac{1}{2}$ piece into two equal parts, each part becomes $\frac{1}{4}$, and two $\frac{1}{4}$ together make $\frac{1}{2}$.

Equivalent Fractions Equivalent fractions are different fractions that represent the same part of a whole. For example:

- $\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8}$
- $\frac{1}{3} = \frac{2}{6} = \frac{3}{9} = \frac{4}{12}$
- $\frac{2}{5} = \frac{4}{10} = \frac{6}{15}$

You can find equivalent fractions by multiplying or dividing both the numerator and denominator by the same number.

Many group activities in the chapter ask you to use a kit or draw shaded shapes to find and list equivalent fractions for $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$, and $\frac{1}{6}$. There are also exercises where you fill in blanks or compare images to find which fractions are the same.

Comparing Fractions—Same Denominator and Numerator

When two fractions have the same denominator (bottom number), the one with the bigger numerator (top number) is larger. For example, between $\frac{1}{3}$ and $\frac{2}{3}$, $\frac{2}{3}$ is bigger. If two fractions have the same numerator, then the one with the smaller denominator is larger. For example, $\frac{1}{5}$ is greater than $\frac{1}{6}$ because if you share something into fewer pieces, each piece is bigger.

A set of practice questions in the chapter help you compare fractions using “greater than” ($>$) and “less than” ($<$) signs. For example:

- $\frac{1}{4} < \frac{3}{4}$
- $\frac{3}{5} < \frac{4}{5}$
- $\frac{5}{7} > \frac{2}{7}$

You are encouraged to use fraction kits and visual models to help with these

comparisons.

Fractions Greater Than One

Fractions can be more than a whole. For example, if you have 5 pieces of $\frac{1}{2}$ paratha, you actually have $\frac{5}{2} = 2$ and $\frac{1}{2}$ parathas (written as $2\frac{1}{2}$). You can show such fractions on a number line by counting equal parts past the number one.

Activity-based questions in the chapter use food items like parathas and pizza to show how fractions greater than one work in real life. For example:

- If each pizza is cut into $\frac{1}{3}$ slices and you eat 4 slices, you have eaten $\frac{4}{3} = 1\frac{1}{3}$ pizzas.
- If you eat 9 pieces of $\frac{1}{4}$ paratha, you have eaten $2\frac{1}{4}$ parathas.

These examples help visualize and understand improper fractions (where the numerator is greater than the denominator) and mixed fractions.

Comparing Fractions with Reference to 1 and $\frac{1}{2}$

Sometimes, you can compare fractions by using 1 or $\frac{1}{2}$ as reference points. For instance, $\frac{7}{8}$ is less than 1, while $\frac{8}{6}$ is more than 1. Similarly, $\frac{3}{6} = \frac{1}{2}$ and $\frac{5}{8}$ is more than $\frac{1}{2}$. Using reference points simplifies comparing fractions, especially when denominators or numerators are not the same.

Various exercises in the chapter help sharpen these comparison skills. You may be asked to circle fractions that are equal to or less than $\frac{1}{2}$, compare lists of fractions, or explain your reasoning using concrete examples.

Practical Activities and Group Work

The chapter also includes hands-on activities like:

- Making different wholes using only pieces of $\frac{1}{6}$ and $\frac{1}{12}$.
- Using number lines to add up several fractions and find the total.
- Solving word problems to apply the concept of fractions in real life, like measuring the length formed by a line of ants if each ant is $\frac{1}{4}$ cm long.

These practical activities reinforce concepts and make learning fractions fun and interactive.

To sum up, the chapter covers how to identify, compare, and find equivalent fractions; work with fractions greater than 1; use visual tools; and solve life-like problems using fraction concepts. Mastery of this chapter is necessary for further topics in mathematics and for real-life applications involving sharing, measuring, and comparing parts.